

Domestic and Feral Cat Gut Microbiomes

Meta Project June 2026

Group 7

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[Link to the github](#)

1 Introduction

The microbiome is a complex compilation of microorganisms that live along and inside organisms. The gut microbiome has a big impact on digestion, immune system, and many other aspects of the host organisms health. The composition of a microbiome varies a lot from species to species and is influenced by a long list of factors such as diet, environment, genetics and even minor differences can lead to very different microbiomes.

This project analyses a global dataset published along with the article, Functional Insights Into the Effect of Feralisation on the Gut Microbiota of Cats Worldwide, where the purpose is to examine how feralization and location can influence the gut microbiome of cats. Our goal is to focus on the taxonomic patterns in the dataset, and if/how we can differentiate between cats by location and origin. Furthermore, we want to find the most prevalent microbial taxa that drives these differences.

2 Understanding the dataset

Fecal samples from feral and domestic cats in six countries, Aruba, Brazil, Cabo Verde, Denmark, Malaysia, and Spain, were collected from veterinary clinics or shelters. The Feral cats were defined as 'individuals that had been originally captured in the streets/nature' and the domesticated cats were defined as 'individuals living with humans prior to entering the clinic or shelter'.

The dataset contains 92 samples collected across six different countries with 15 to 17 samples per country. See table 1¹

Location	Domestic	Feral	Total
Aruba	9	8	17
Brazil	8	8	16
CaboVerde	12	3	15
Denmark	6	9	15
Malaysia	8	7	15
Spain	8	7	15

Table 1: Domestic and Feral cat samples per country

For the MAGs assembled, a total of 214 taxa were identified across the 92 samples. The lengths of the assembled MAGs range from 966643 to 4757748 nucleotides, with most genomes being between 1 and 3 million basepairs long. See figure 1.

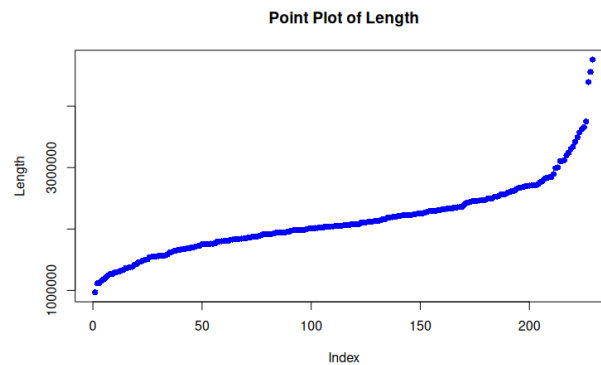


Figure 1: Lengths of MAGs by index

The dataset does not contain raw abundance measurements, making it difficult to tell whether one sample contains more bacteria in comparison to another sample. Instead the dataset contains the relative abundance, which describes the proportion of each bacteria within a sample. The relative abundances are plotted below.

¹All figures presented in this report can be found in the Group7.Rmd file located in the Final_project folder in the main branch of our Git repository.

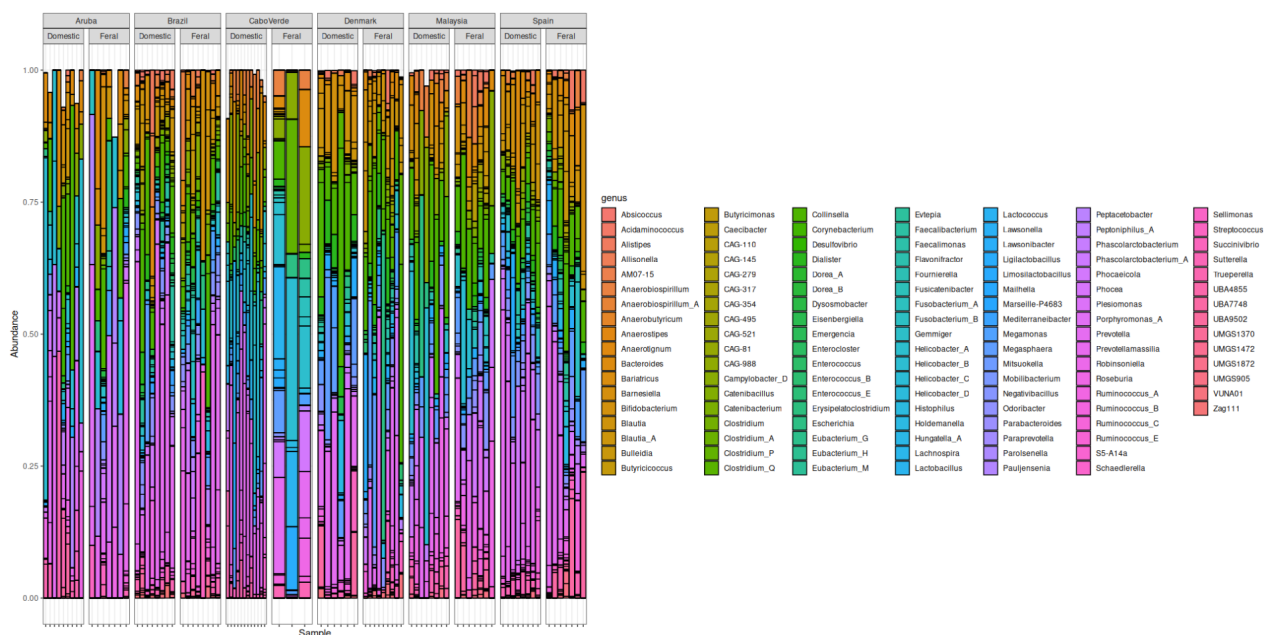


Figure 2: Relative abundance for the samples

Several observations can be made from figure 2. The samples from Aruba contains the least assembled MAGs compared to the other locations. This might suggest the presence of bacteria that were not found in the reference libraries and therefore not successfully reconstructed into MAGs.

The feral samples from Cabo Verde look noticeably dominated by fewer bacteria compared to the other samples. Overall, no other noticeable differences between locations or between origins of the cats are clear from this figure.

Regarding the quality of each bin, we notice a high completeness and in general low contamination. See figure below.

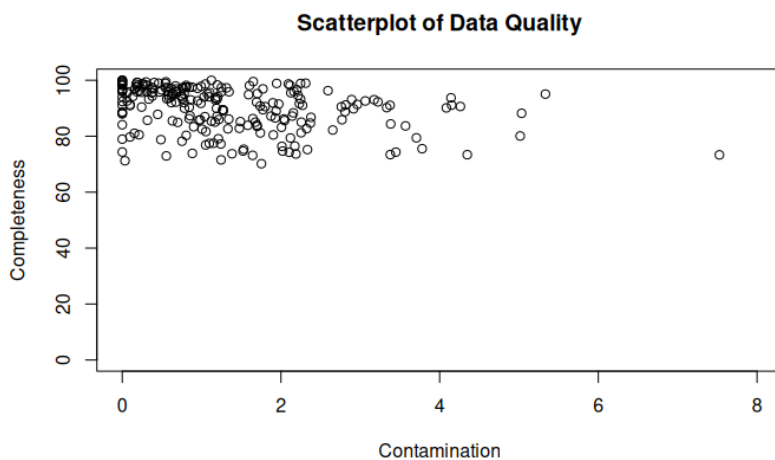


Figure 3: Contamination vs Completeness

There are a few points in Figure 3 that differ slightly in quality from the rest of the MAGs. The most notable MAG, located at approximately 8% contamination and 70% completeness, could potentially be considered for removal. However, it still falls within the moderate-confidence quality standards and is therefore not removed.

We also notice that the completeness never goes below 70% and no contamination is above 8%. These values might have been used as filtering thresholds for trimming during the preprocessing, or at least suggest similar quality criteria were applied.

3 Analysis

3.1 Alpha diversity

Having established that the dataset is of sufficient quality we can now begin exploring the taxa and how microbial communities differ within samples and between groups. Alpha diversity describes the diversity within individual samples. By comparing alpha diversity values across groups, such as location or origin, we can assess whether cats from different locations or origins tend to differ in microbial richness and diversity.

First, we calculated alpha diversity for each individual sample using three different tests: Shannon diversity, Observed Richness, and Phylogenetic Diversity. These tests capture different aspects of diversity within samples, such as richness, evenness, and phylogenetic diversity. To determine whether alpha diversity differed between groups, we performed Kruskal–Wallis tests using both location and cat origin as grouping variables. We considered the results statistically significant when $p\text{-value} < 0.05$, corresponding to a confidence level of 95%, and compiled the results in the following table.

Method	Chi-squared	$p\text{-value}$	df
Shannon location	22.28	$4.64 \cdot 10^{-4}$	5
Observed Richness location	44.93	$1.50 \cdot 10^{-8}$	5
Phylogenetic Diversity location	33.13	$1.42 \cdot 10^{-6}$	5
Shannon origin	1.15	0.28	1
Observed Richness origin	0.54	0.46	1
Phylogentic Diversity origin	0.20	0.65	1

Table 2: Kruskal-Wallis based on method and variable

It is clear from Table 2 that the location has a significant effect on microbiome diversity between cats, whereas origin does not. All the location $p\text{-values}$ are well below 0.05, while the origin $p\text{-values}$ are considerably higher than the significance level. This indicates that location explains more of the observed variation in alpha diversity than origin does.

Now that we know that location is the significant sample variable, we will continue to explore the dataset with this using this variable in the following tests. To further explore the differences in alpha diversity between locations, and to compare the different diversity test,

boxplots were chosen to visualize each alpha diversity measure and are shown in the figure below.

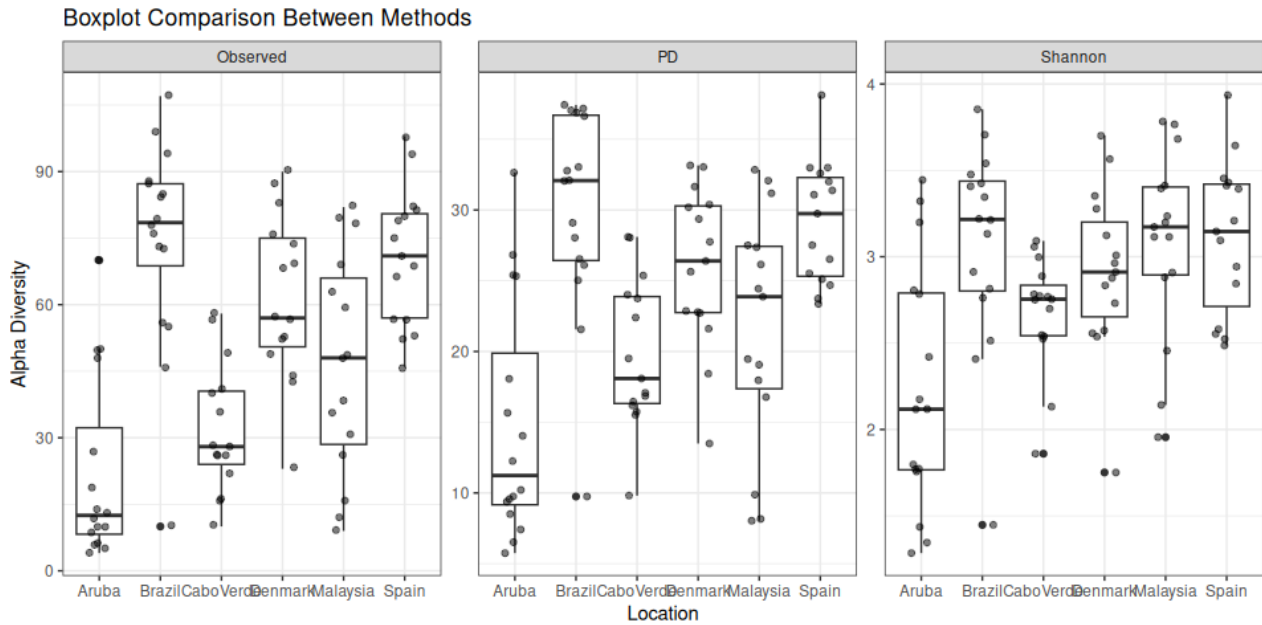


Figure 4: Boxplot Comparison Between Methods

In general the different methods for alpha diversity concludes roughly the same results. There is a discrepancy of what samples are chosen as outliers between the different methods, but in general they tell the same story. The alpha diversity is consistently lower for Aruba and Cabo Verde and higher for the rest of the locations.

There are 3 samples from Aruba and 2 from Brazil that are under the expected 1.5 barrier for the Shannon index. The samples from Aruba are in general lower in diversity and do not look like outliers, but more like a result of the general environment that the cats live in. The Brazillian samples however, are way lower then the rest of the samples which might suggest bad quality of the 2 samples.

3.2 Beta diversity

While alpha diversity provides information about the diversity within individual samples, it does not indicate whether the microbial communities differ between groups. To investigate this we therefore applied a beta diversity analysis. We calculated beta diversity using four different methods: Bray-Curtis distance, Jaccard distance, unweighted and weighted UniFrac. These methods capture different aspects of diversity across groups, such as relative abundance, presence or absence of taxa, and phylogenetic relations. Therefore, we wanted to show the differences between the methods.

As we saw in the alpha diversity analysis, origin is not very significant, and also when applying these different methods to compare origin, we observe roughly the same results. See dispersion plot Figure 5

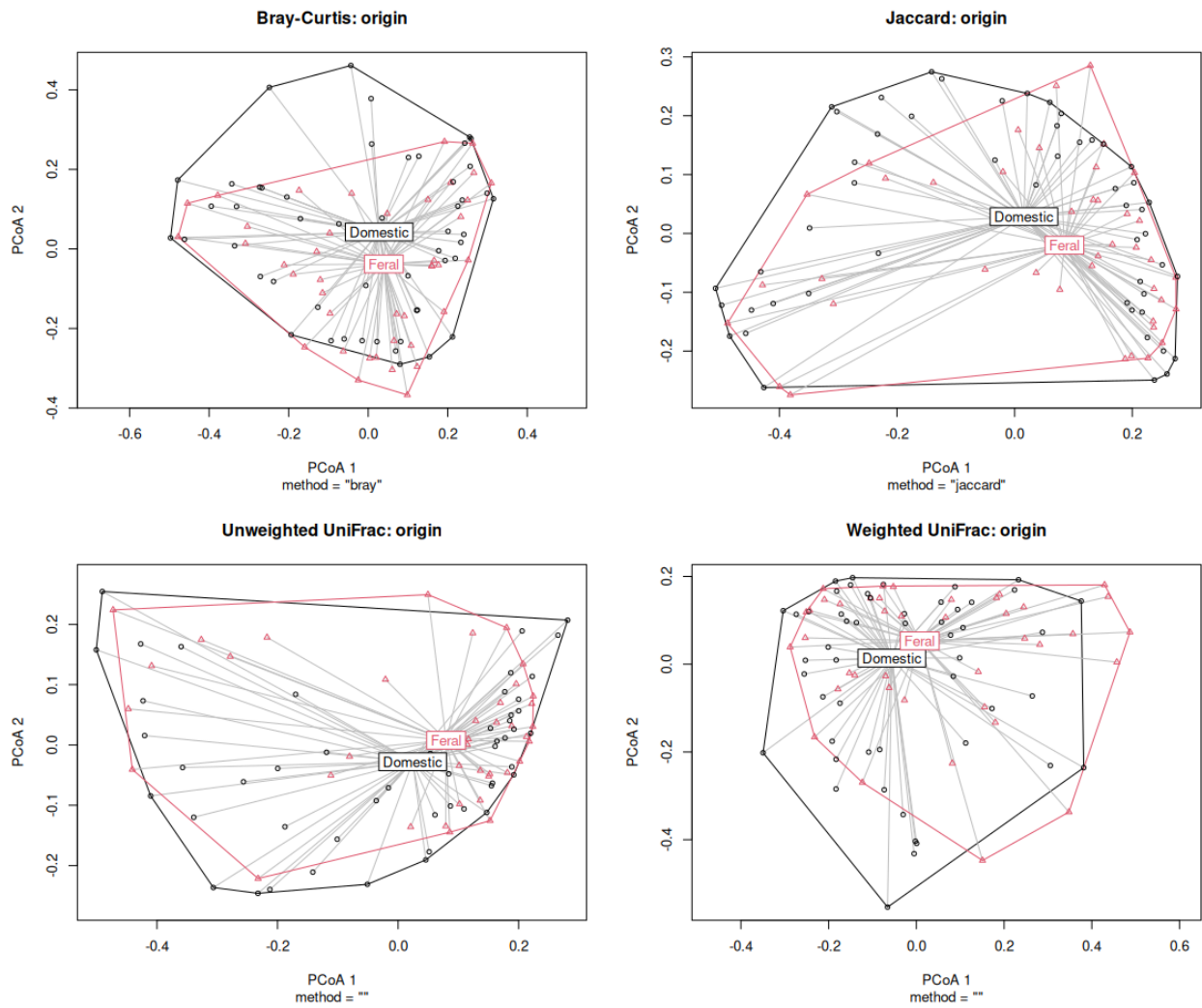


Figure 5: Dispersion plots by origin

When we apply these methods to compare location we get more meaningful dispersion plots:

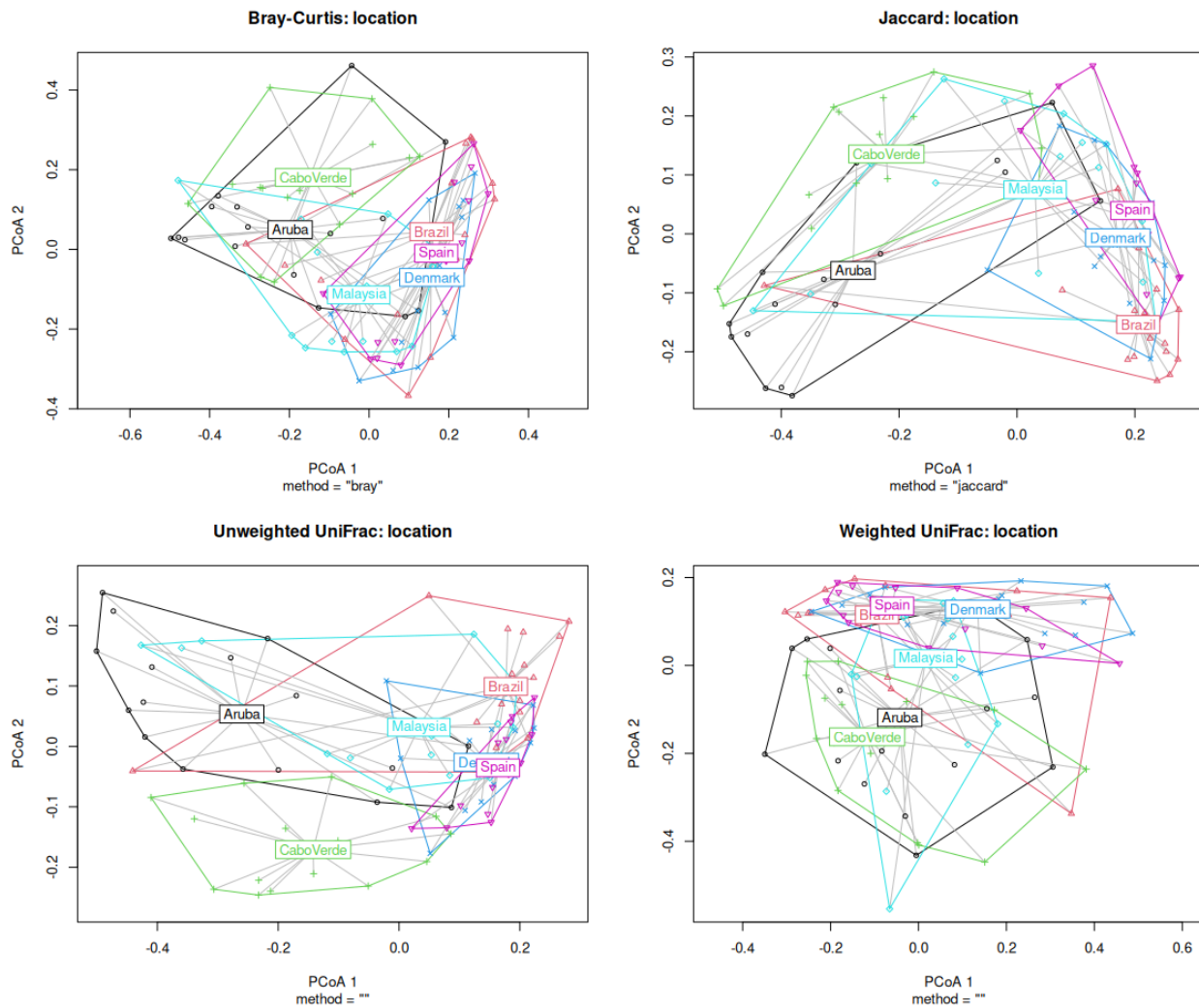


Figure 6: Dispersion plots by location

We can gather a couple of things from figure 6. Firstly, the Aruba and Cabo Verde profiles differ from the other locations. It could also be argued that Malaysia has a more distinct profile compared to Denmark, Spain and Brazil.

Second, the Danish, Spanish, and Brazilian samples are in general more compact and behave more like clusters compared to the bigger spread of Cabo Verde and Aruba. This suggests that these locations are more similar and/or more meaningful for the building of the PCoA. Another interesting observation is the high spread of Aruba and Cabo Verde considering these were within the lowest alpha diversity in figure 4.

When analysing these dispersions, we also have to take the PCoA weights into account. See Table 3

Method	PCoA 1	PCoA 2	Total amount explained
Bray-Curtis	15.7%	11.5%	27.2%
Jaccard	21.3%	7.6%	28.9%
Unweighted Unifrac	30.0%	8.5%	38.5%
Weighted Unifrac	27.1%	20.8%	47.9%

Table 3: PCoA weights

The Weighted UniFrac test explained almost 50% of the data, which is a rather large amount and therefore a better representation of how the actual data looks.

Since we are handling complex data with many different bacteria, it makes sense to apply a PERMANOVA instead of a traditional ANOVA. This converts the raw data to a distance matrix instead of single response variables. The resulting p -values from all the PERMANOVA tests were less than 0.001 when a test was based on location, and above 0.05 when based on origin.

Because of the observations from Figure 6, we speculate that Brazil, Spain, and Denmark shares similarities, as do Cabo Verde and Aruba. We therefore decided to investigate whether these observations were statistically significant.

For these analyses, we used the same four beta diversity measures, but performed pairwise comparisons between each locations, resulting in the matrices below:

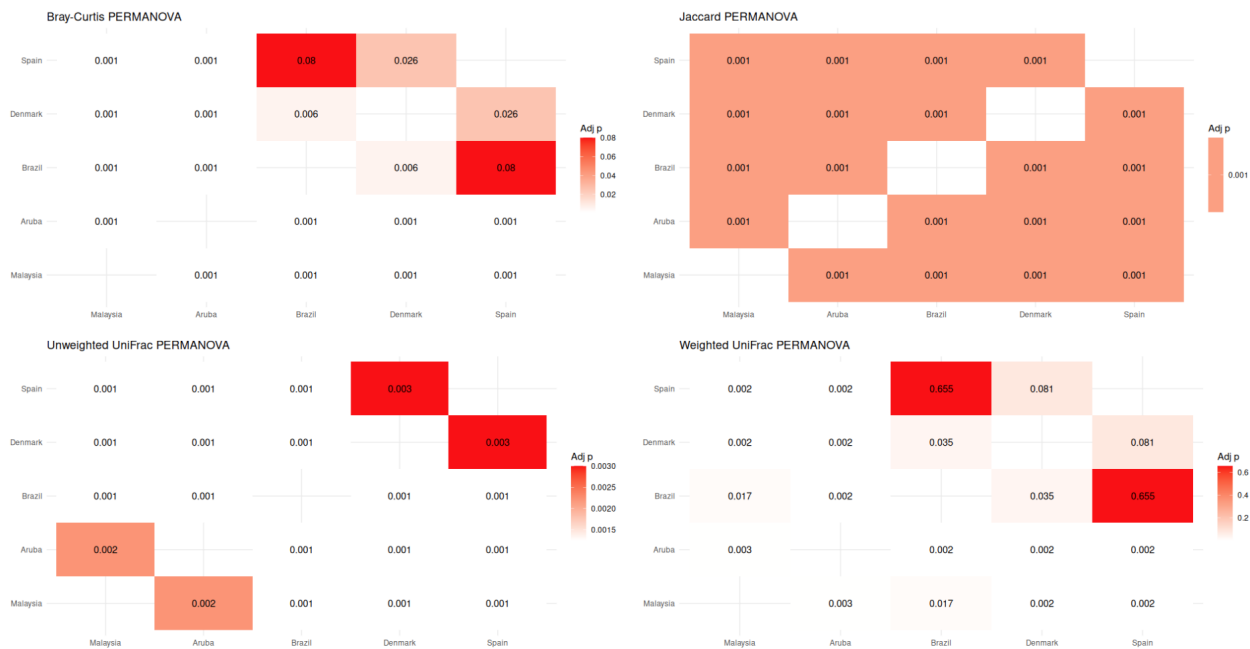


Figure 7: PERMANOVA p-tests

Figure 7 indeed tells us about some of the relationships between the locations. Jaccard distance and unweighted UniFrac only found statistically significant differences between locations, suggesting that presence/absence information alone, is not enough to capture the diversity. On the contrary, Bray-Curtis and weighted UniFrac, identified a significant relationship between Brazil and Spain. In addition weighted UniFrac also indicates a closer

relationship between Spanish and Danish samples with a p -value of 0.081, but Bray-Curtis did find this relation significant with a p -value of 0.026.

3.3 Differential abundance

Our beta diversity analysis demonstrated that the microbial community differed more strongly based on the location rather than the origin of the cats. However, beta diversity analyses only concludes that the communities differ overall, but does not indicate what microbial taxa contribute to the differences. To investigate the potential drivers of the microbial diversity, differential abundance analyses were performed. These focus on the individual taxa and determine whether specific microorganisms were more enriched in one group relative to others.

Before performing the analyses, it is important to explore the distribution of the taxa across the samples, as microbiome datasets typically contain many rare taxa that are present in only a small number of samples. As a result, we often see a large number of zero observations.

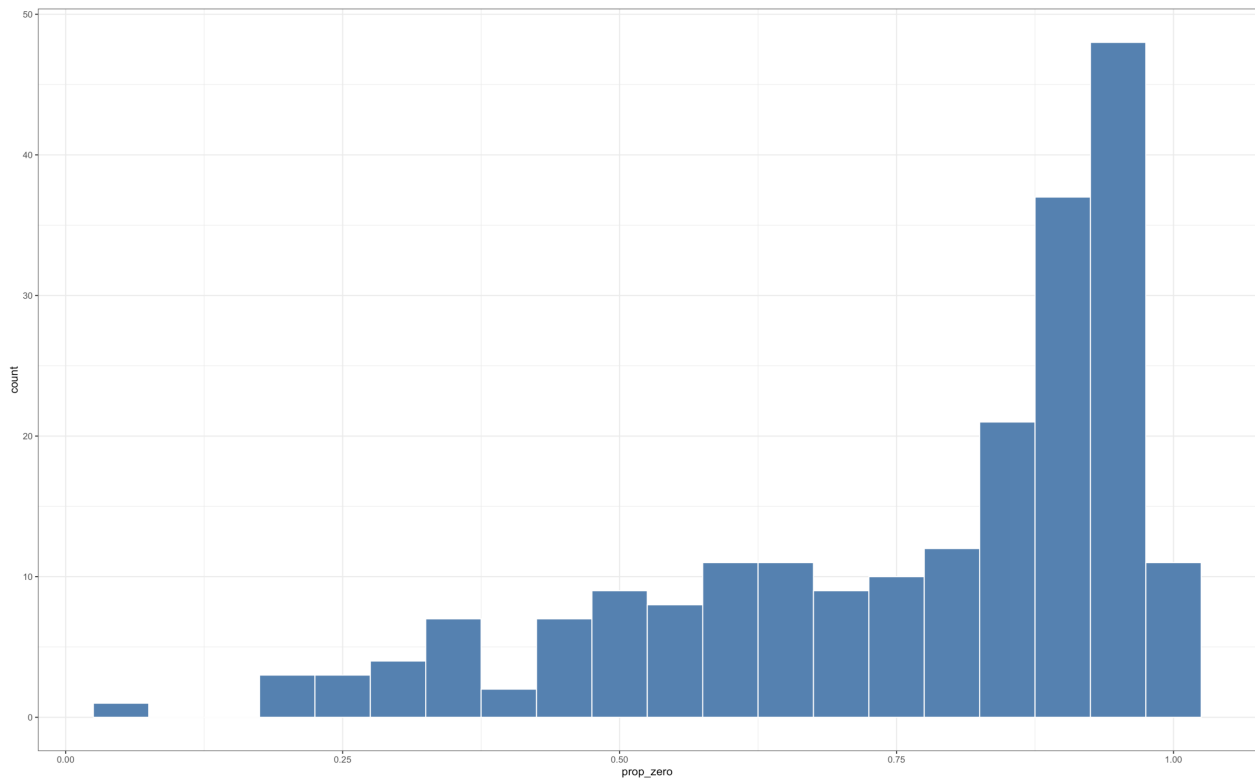


Figure 8: Zero inflation plot

Figure 8 shows a zero inflation plot. While alpha and beta diversity describes the overall differences in the microbial community, they do not identify the specific taxa that contributes to the minor divergences. For that reason, we have constructed volcano plots to understand which taxa are associated with specific locations.

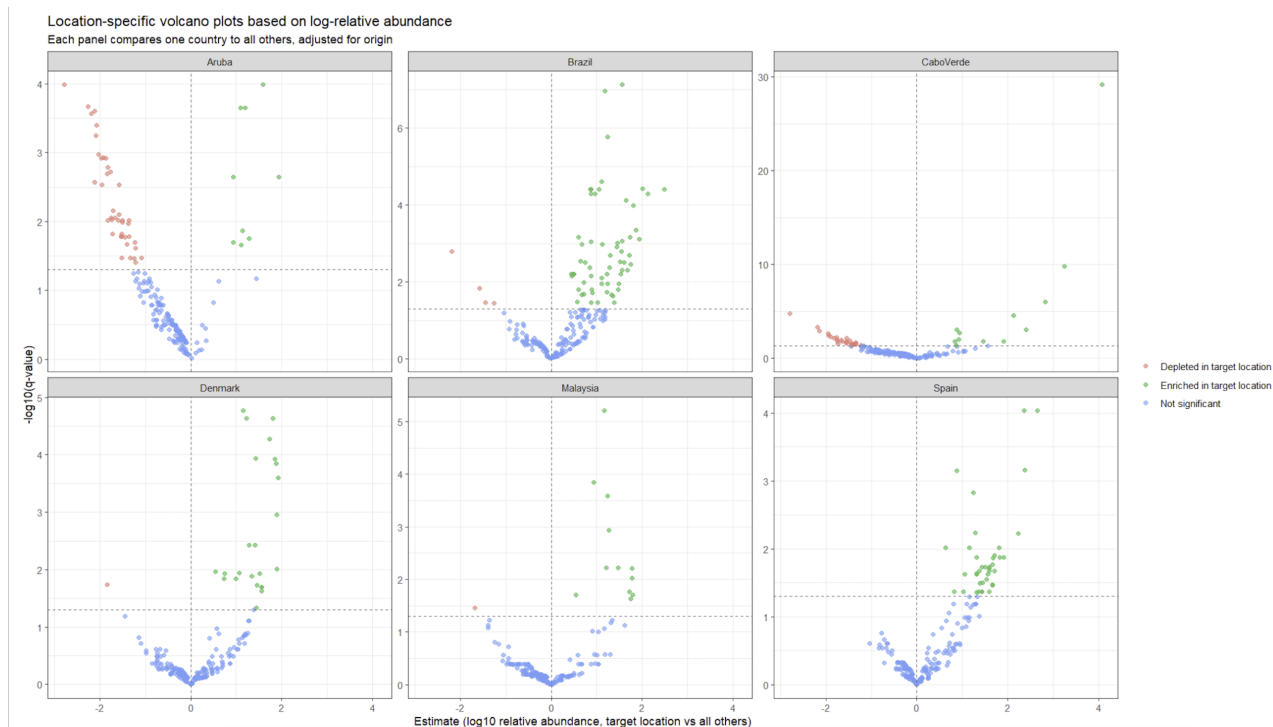


Figure 9: Volcano plot

Volcano plots visualize effect size and statistical significance. An interesting observation is, that we see an association between Aruba and Cabo Verde. These two have a common denominator being samples taken from islands. This is a similar result to our beta diversity where, the two locations appeared distinct from the mainland samples.

While the volcano plot identifies location specific taxa, they do not address whether domestication contributed to microbiome composition independently of location. To investigate whether domestication contributes to the gut microbiome variety, we performed a redundancy analysis.

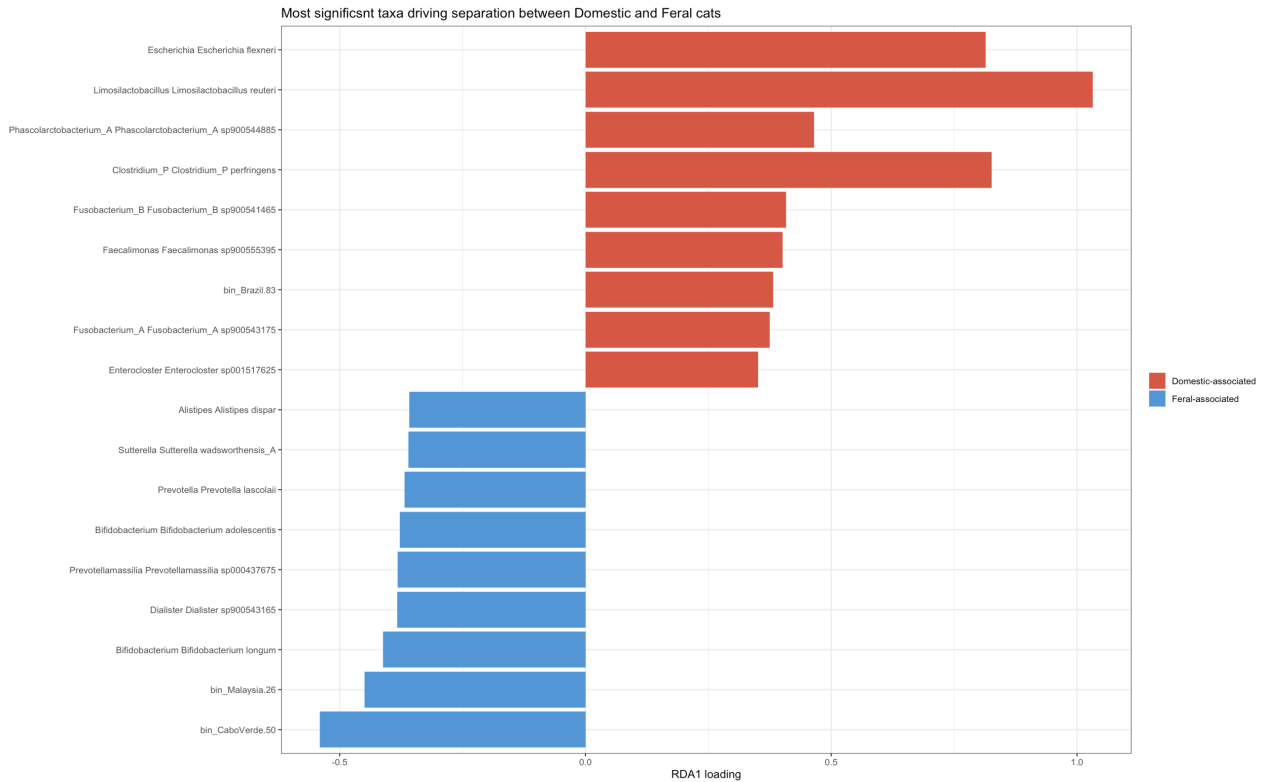


Figure 10: RDA

Figure 10 shows the RDA1 loadings for the taxa that most strongly separate domestic from feral cats, after controlling for location. Positive loadings (red bars) indicate association with domestic cats; negative loadings (blue bars) indicate association with feral cats. Nine taxa have positive loadings, with the highest reaching approximately +1.0, suggesting a stronger signal in the domestic direction. Nine taxa have negative loadings, but their absolute values are smaller, with the most negative reaching only about -0.5 . This asymmetry indicates that the domestication effect is driven more by a set of taxa enriched in domestic cats than by taxa enriched in feral cats.

4 Discussion

The consensus throughout this project was that the difference in the microbiome was primarily driven by location. We observed that locations with island ecosystems have their own separate clusters compared to the mainland countries. This indicates that the isolated environments like Aruba and Cabo Verde offer a smaller but more unique microbial diversity. Furthermore, we saw statistical similarities between the samples from Denmark and Spain. This could be a result of being on the same continent and higher migration between the two countries.

A surprising observation was the similarity between Spain and Brazil. In spite of their long distance from each other, Brazil and Spain still managed to overlap through our different analyses. One possible explanation is the historical and cultural connection between

South America and the south of Europe. This might have indirectly influenced cat populations through shared human lifestyles, food sources and pet traditions. Brazil also contains one of the worlds largest domestic cat populations with about 13.3 million cats, but there is estimated to be about 22.1 million cats in all of Brazil.² The migration of people and animals between these countries may have facilitated the exchange of microbial communities.³ An important consideration when interpreting the results is the classification of cats as "Domestic" or "Feral". While a binary classification proves to be practical in many analyses, it may simplify the reality entirely. In the original study, all cats had to go through behavioural observations to "prove" their domestication, where feral cats often displayed a more reserved and violent behaviour. The study did not have any guidelines in regards to domestic cats being able to infiltrate the environment of feral cats. Interactions between these 2 groups could have lead to a mix of microbiomes.

5 Conclusion

The purpose of this project was to investigate whether feralisation influences the gut microbiome of cats and to determine the importance of domestication and location for microbial communities. The microbial analyses revealed differences between locations, whereas no significant differences was detected between the cats origin. Cats from Aruba and Cabo Verde generally exhibited lower diversity than cats from the mainland locations, suggesting that local environment influences richness and phylogenetic diversity. Overall, the results do not prove the hypothesis that origin drives the variation of the gut microbiome of cats. Rather it was the location that proved to be the most influential factor across our analytical tests. While domestication appears to contribute to the microbiome in some way, it does not account for the strong location-dependent observations.

Taken together, these analyses suggest environmental and geographic factors play a larger role in shaping the gut microbiome than domestication alone.

²<https://worldpopulationreview.com/country-rankings/cat-population-by-country>

³<https://www.icmpd.org/file/download/54178/file/Brazil.pdf>, page 150, table 8